

Differential Geometry

Exercise 3

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3.1 Problem

Let M be a set and $\mathcal{A} = \{(\mathbf{x}_\alpha, U_\alpha)\}_\alpha$ an atlas on M . For α, β define the map

$$\mathbf{x}_{\alpha\beta}: \mathbf{x}_\alpha^{-1}(\mathbf{x}_\beta(U_\beta)) \rightarrow U_\alpha \times U_\beta, \quad x_{\alpha\beta}(u) = (u, \mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha(u)).$$

- (1) Show that if $\mathbf{x}_{\alpha\beta}$ is always proper, then the induced topology on M is Hausdorff.
- (2) Show that for $BK = \mathbb{R}, \mathbb{C}, \mathbb{K}P^n$ is Hausdorff.

- (1) We can view M as the quotient manifold of $\coprod_\alpha U_\alpha$ under the equivalence that $p \in U_\alpha$ and $q \in U_\beta$ are identified when $\mathbf{x}_\alpha(p) = \mathbf{x}_\beta(q)$. If we are given two distinct points $p \in U_\alpha, q \in U_\beta$ then (p, q) cannot be in the image of $\mathbf{x}_{\alpha\beta}$ as that would imply that $\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha(p) = q$, i.e. that $\mathbf{x}_\beta(q) = \mathbf{x}_\alpha(p)$, a contradiction. $\mathbf{x}_{\alpha\beta}$ is proper and thus closed (since all spaces in question are locally compact Hausdorff), and so its image is closed in $U_\alpha \times U_\beta$. Thus there exist open sets $p \in V_\alpha \subseteq U_\alpha$ and $q \in V_\beta \subseteq U_\beta$ which don't intersect $\mathbf{x}_{\alpha\beta}$'s image. These are open in M , and they must be disjoint: if $r \in V_\alpha, r' \in V_\beta$ are such that $\mathbf{x}_\alpha(r) = \mathbf{x}_\beta(r')$ then r is in $\mathbf{x}_{\alpha\beta}$'s domain and

$$\mathbf{x}_{\alpha\beta}(r) = (r, \mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha(r)) = (r, r') \in V_\alpha \times V_\beta,$$

a contradiction to $V_\alpha \times V_\beta$ not intersecting $\mathbf{x}_{\alpha\beta}$'s image.

- (2) The charts of $\mathbb{K}P^n$ are given by $\mathbf{x}_i: \mathbb{K}^n \rightarrow \mathbb{K}P^n$ where $\mathbf{x}_i(v)$ inserts 1 at the i th index of v (and takes the equivalence). Its image is the set of classes of the form $[v]$ where $v_i \neq 0$, let this be U_i . Its inverse is $\mathbf{x}_i^{-1}: U_i \rightarrow \mathbb{K}^n$ given by $\mathbf{x}_i^{-1}([v]) = 1/v^i \hat{v}^i$ where $\hat{v}^i$ denotes the removal of the i th index.

The maps $\mathbf{x}_{ij}$ then are

$$\mathbf{x}_{ij}: \mathbf{x}_i^{-1}(U_j) \rightarrow U_i \times U_j, \quad v \mapsto (v, \mathbf{x}_j^{-1} \circ \mathbf{x}_i(v)).$$

$\mathbf{x}_j^{-1} \circ \mathbf{x}_i(v)$ is equal to $1/v^j(v_1, \dots, \hat{v}_j, \dots, v_{i-1}, 1, v_i, \dots, v_n)$ when $j < i$ (for $j > i$ the indices are ordered a bit differently; $\hat{v}_j$ denotes the removal of v_j). Now, $\mathbf{x}_i^{-1}(U_j)$ is the set of all $v \in \mathbb{K}^n$ such that $\mathbf{x}_i(v)$'s j th index is nonzero, this is just the set of vectors in $\mathbb{K}^n$ whose j th index is nonzero. So $\mathbf{x}_{ij}$'s image is all (v, u) such that $\mathbf{x}_i(v) = \mathbf{x}_j(u)$. This occurs iff $(v_1, \dots, 1, \dots, v_n)$ and $(u_1, \dots, 1, \dots, u_n)$ are linearly dependent: $\lambda(v_1, \dots, 1, \dots, v_n) = (u_1, \dots, 1, \dots, u_n)$ for some λ .

Let us denote insertion of 1 at index i of y_i , so $\mathbf{x}_{ij}$'s image is

$$\text{im } \mathbf{x}_{ij} = \{(v, u) \in U_i \times U_j \mid \lambda y_i(v) = y_j(u) \text{ for some } \lambda \in \mathbb{K}\}.$$

If $i < j$ then λ must be u_i so that the image is

$$\{(v, u) \mid u_i y_i(v) = y_j(u)\}$$

which is the preimage of zero of the continuous map $(v, u) \mapsto u_i y_i(v) - y_j(u)$. Thus the image is closed. Similar for $i > j$, and when $i = j$ the image is just the diagonal which is again closed.

We showed above that $\mathbf{x}_{\alpha\beta}$'s image being closed is enough to ensure the manifold being Hausdorff.

3.2 Problem

Let $\text{Mat}_n(\mathbb{C})$ denote the space of $n \times n$ complex matrices. Denote by $M^\dagger$ the conjugate transpose of $M \in \text{Mat}_n(\mathbb{C})$. A matrix M is **unitary** if $M^\dagger M = I$, and the group of unitary matrices is denoted U_n . A matrix M is **skew-Hermitian** if $M^\dagger = -M$, and denote by $\mathfrak{u}_n$ the space of skew-Hermitian matrices. We will show that U_n is a Lie group and we can identify its Lie algebra with $\mathfrak{u}_n$.

- (1) Let $\text{Herm}_n \subseteq \text{Mat}_n(\mathbb{C})$ be the space of Hermitian matrices $M = M^\dagger$. Use the obvious maps $\mathbb{C}^{n^2} \rightarrow \text{Mat}_n(\mathbb{C})$ and $\mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n \rightarrow \text{Herm}_n$ to define smooth structures. Define $F: \text{Mat}_n(\mathbb{C}) \rightarrow \text{Herm}_n$ by $F(M) = M^\dagger M$. Show that the identity I is a regular point of F .
- (2) Use the invariance of F under left multiplication by unitary matrices to show that all $U_n \subseteq \text{Mat}_n(\mathbb{C})$ are regular points of F .
- (3) Conclude that $I \in \text{Herm}_n$ is a regular value of F , and thus U_n is an embedded submanifold of $\text{Mat}_n(\mathbb{C})$ of dimension n^2 .
- (4) Let $P \subseteq M$ and $Q \subseteq N$ be embedded submanifolds, and $F: M \rightarrow N$ smooth with $F(P) \subseteq Q$. Show that $F|_P: P \rightarrow Q$ is smooth.
- (5) Show that multiplication $U_n \times U_n \rightarrow U_n$ and inversion $U_n \rightarrow U_n$ are smooth, and thus U_n is a Lie group.
- (6) Let $\iota: U_n \rightarrow \mathbb{C}^{n^2}$ be the inclusion map. Show that the composition

$$T_e U_n \xrightarrow{d\iota_e} T_e \mathbb{C}^{n^2} \xrightarrow{\zeta} \mathbb{C}^{n^2}$$

induces an isomorphism of vector spaces between $T_e U_n$ and $\mathfrak{u}_n$.

- (1) The map $\mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n \rightarrow \text{Herm}_n$ is given by mapping $(z_i) \oplus (x_i)$ to the Hermitian matrix whose diagonal is given by the x_i s, above the diagonal by z_i s, and below by $\bar{z}_i$ s. This is a real linear isomorphism, in particular a bijection, and thus induces a smooth structure. Let $\Phi: \mathbb{C}^{n^2} \rightarrow \text{Mat}_n(\mathbb{C})$ be the obvious isomorphism and $\Psi: \text{Herm}_n \rightarrow \mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n$ the inverse of the above bijection.

Then the coordinate representation of F maps $M \in \mathbb{C}^{n^2}$ to the upper triangle of $M^\dagger M$ and its diagonal. We have that

$$(M^\dagger M)_{ij} = \sum_t \bar{M}_{ti} M_{tj},$$

and each of these coordinates is smooth over $\mathbb{R}$ (complex multiplication is smooth, and so is conjugation as it just the map $(x, y) \mapsto (x, -y)$). Thus all of F 's coordinates are smooth, and so F is.

- (2) For a unitary matrix U , we have that

$$F(UM) = M^\dagger U^\dagger U M = M^\dagger M = F(M),$$

and since the action of matrix multiplication in $\text{Mat}_n(\mathbb{C})$ is a smooth action by $\text{GL}_n(\mathbb{C})$, we have that left multiplication by $U \in U_n \subseteq \text{GL}_n(\mathbb{C})$ is a diffeomorphism of $\text{Mat}_n(\mathbb{C})$. Thus (denoting left multiplication under U by U), we have that for all $M \in \text{Mat}_n(\mathbb{C})$,

$$d(F \circ U)_M = dF_{UM} \circ dU_M = dF_M.$$

Since U is a diffeomorphism and thus of full rank, dF_{UM} and dF_M must have the same rank for all $U \in U_n, M \in \text{Mat}_n(\mathbb{C})$. In particular if U, Q are unitary then so is UQ^{-1} , and thus $dF_{UQ^{-1}Q} = dF_U$ and dF_Q have the same rank.

In particular, I is unitary and so for $U \in U_n$, dF_U and dF_I have the same rank, which we showed is full. Thus every $U \in U_n$ is a regular point of F .

- (3) $I \in \text{Herm}_n$'s level set consists of all unitary matrices, which are all regular points. Thus I is a regular value, and its level set U_n is an embedded submanifold of $\text{Mat}_n(\mathbb{C})$. F 's rank at $U \in U_n$ is full, thus equal to $\dim \mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n = n(n-1) + n = n^2$. This means U_n 's codimension is n^2 , and thus its dimension is $2n^2 - n^2 = n^2$.
- (4) Since $P \subseteq M$ is a submanifold, the inclusion $\iota_P: P \hookrightarrow M$ is smooth. $F|_P: P \rightarrow M$ is just $F \circ \iota_P$ and thus is smooth.

Now we want to show that we can restrict the codomain. Let F' denote the restriction on the codomain $F' = F|_P: P \rightarrow Q$; we note that it is still continuous. Since $\iota_Q: Q \hookrightarrow N$ is a smooth embedding, and $\iota_Q \circ F' = F$. Since ι_Q is in particular a smooth immersion and thus of constant rank, at every point of Q there is a chart (U, ψ) and a chart $(V, \tilde{\psi})$ of N such that $U \subseteq V$, and the coordinate representation is of the form

$$\tilde{\psi}^{-1} \circ \iota_Q \circ \psi: \bar{x} \mapsto (\bar{x}, 0).$$

If we denote by π the projection $(\bar{x}, \bar{y}) \mapsto \bar{x}$ we see that

$$\pi \circ \tilde{\psi}^{-1} \circ \iota_Q \circ \psi = \text{id} \implies \psi^{-1} = \pi \circ \tilde{\psi}^{-1} \circ \iota_Q.$$

Then for $p \in P$ we can choose coordinate charts $\psi \subseteq Q$ and $\tilde{\psi} \subseteq N$ such that $F(p)$ is in ψ (and thus $\tilde{\psi}$'s) domain. We then choose a coordinate chart $\varphi \subseteq P$ around p which is mapped by F' into ψ 's domain (since F' is continuous, its preimage of ψ 's domain is open). Then F' 's coordinate representation is

$$\psi^{-1} \circ F' \circ \varphi = \pi \circ \tilde{\psi}^{-1} \circ \iota_Q \circ F' \circ \varphi = \pi \circ (\tilde{\psi}^{-1} \circ F \circ \varphi).$$

Since F is smooth, $\tilde{\psi}^{-1} \circ F \circ \varphi$ is smooth, and the projection π is smooth as well. Thus $\psi^{-1} \circ F' \circ \varphi$ is smooth, meaning F' has a smooth coordinate representation in some neighborhood of any point $p \in P$, so F' is smooth as desired.

- (5) $\text{GL}_n(\mathbb{C}) \subseteq \text{Mat}_n(\mathbb{C})$ and $U_n \subseteq \text{Mat}_n(\mathbb{C})$ are embedded submanifolds. This means that U_n is an embedded submanifold of $\text{GL}_n(\mathbb{C})$: its inclusion in $\text{Mat}_n(\mathbb{C})$ is the composition of inclusions $U_n \hookrightarrow \text{GL}_n(\mathbb{C}) \hookrightarrow \text{Mat}_n(\mathbb{C})$. This composition is a smooth immersion, and thus the first inclusion $U_n \hookrightarrow \text{GL}_n(\mathbb{C})$ must be a smooth immersion as well; thus $U_n \subseteq \text{GL}_n(\mathbb{C})$ is embedded. Thus we can restrict multiplication and inversion of $\text{GL}_n(\mathbb{C})$ to give the multiplication and inversion of U_n , and these remain smooth.
- (6) The dimension of $\mathfrak{u}_n$ is n^2 , as a skew-Hermitian matrix is determined by the imaginary values on its diagonal, and its complex upper triangle. That is, there is an obvious isomorphism between it and $\mathbb{C}^{n(n-1)/2} \oplus (i\mathbb{R})^n$, which has dimension n^2 . Thus $T_e U_n$ and $\mathfrak{u}_n \subseteq \mathbb{C}^{n^2}$ have the same dimension. Since $\iota: U_n \rightarrow \text{Mat}_n(\mathbb{C})$ is a smooth immersion, $d\iota_e$ is injective. Thus we need only show that the composition lands in $\mathfrak{u}_n$, as then we have an injective map $T_e U_n \rightarrow \mathfrak{u}_n$ and since they are of the same dimension, it must be an isomorphism.

Let $\gamma: (-\varepsilon, \varepsilon) \rightarrow U_n$ be smooth and centered at I , then $\iota \circ \gamma$ is a smooth curve which lands in $U_n \subseteq \text{Mat}_n(\mathbb{C})$. Identify γ with $\iota \circ \gamma$; we want to show that if $\gamma(t) \in U_n$ for all t then $\gamma'(0) \in \mathfrak{u}_n$. Notice that the Hermitian transpose is smooth (over $\mathbb{R}$) whose differential is itself. Since $\gamma(t)\gamma(t)^\dagger = I$, differentiating both sides at zero gives

$$0 = \gamma(0)\gamma'(0)^\dagger + \gamma'(0)\gamma(0)^\dagger = \gamma'(0)^\dagger + \gamma'(0),$$

meaning $\gamma'(0)$ is indeed skew-Hermitian, as desired.

3.3 Problem

For $M = \mathbb{R}^n$, let $r: M \rightarrow M$ be a smooth retraction, meaning $r \circ r = r$.

- (1) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be smooth with a differential of constant rank k . Show that there is a coordinate frame $0 \in (U, \mathbf{x})$ on $\mathbb{R}^n$ such that

$$f \circ \mathbf{x}(x_1, \dots, x_n) = f \circ \mathbf{x}(x_1, \dots, x_k, 0, \dots, 0).$$

- (2) Let $T \subseteq \mathbb{R}^n$ be connected and for $t \in T$ define $A(t): \mathbb{R}^n \rightarrow \mathbb{R}^n$ so that $t \mapsto A(t)$ is continuous. Further suppose that $A(t)$ is a projection for all t , $A(t)^2 = A(t)$. Show that the rank of the matrix $A(t)$ must be constant.
- (3) Show that there is a neighborhood Q of $\text{im} r$ such that dr_q has constant rank for $q \in Q$.
- (4) Show that for each $q \in \text{im} r$ there is a coordinate system $(U, \mathbf{x})$ on M centered at q such that $r(\mathbf{x}(U)) \subseteq \mathbf{x}(U)$ and

$$r \circ \mathbf{x}(x_1, \dots, x_n) = r \circ \mathbf{x}(x_1, \dots, x_k, 0, \dots, 0).$$

- (5) Show that $r(M) \cap \mathbf{x}(U) = r(\mathbf{x}(U)) \cap \mathbf{x}(U)$
- (6) Show that $r(M)$ is an embedded submanifold.

- (1) This proof is essentially the same as that of the first part of the rank theorem. Let us write $F(x, y) = (Q(x, y), R(x, y))$ for $Q: \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $R: \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$. Then F 's Jacobian is

$$JF = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j} & \frac{\partial Q^i}{\partial y^j} \\ \frac{\partial R^i}{\partial x^j} & \frac{\partial R^i}{\partial y^j} \end{pmatrix}.$$

Since it has rank k , it has a $k \times k$ invertible submatrix, which we can assume without loss of generality is the upper left. We can assume this via a change of coordinates, and noting that by continuity if a submatrix has nonzero determinant, it has nonzero determinant in an open neighborhood of the point.

Let us then define $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\varphi(x, y) = (Q(x, y), y)$. Its Jacobian is

$$J\varphi = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j} & \frac{\partial Q^i}{\partial y^j} \\ 0 & \delta_{ij} \end{pmatrix},$$

which is invertible (in the appropriate neighborhood). Thus by the inverse function theorem has an inverse $\varphi^{-1}(x, y) = (\tilde{Q}(x, y), \tilde{R}(x, y))$, and we see then that comparing $\varphi \circ \varphi^{-1}, \varphi^{-1} \circ \varphi$ with the identity, we have $\tilde{R}(x, y) = y$ and that $Q(\tilde{Q}(x, y), y) = x$ and $\tilde{Q}(Q(x, y), y) = x$.

Now consider $F \circ \varphi^{-1}$, which is equal to

$$F \circ \varphi^{-1}(x, y) = (x, R(\tilde{Q}(x, y), y)) = (x, A(x, y))$$

and has Jacobian

$$J(F \circ \varphi^{-1})(x, y) = \begin{pmatrix} \delta_{ij} & 0 \\ \frac{\partial A^i}{\partial x^j} & \frac{\partial A^i}{\partial y^j} \end{pmatrix}.$$

φ^{-1} is a diffeomorphism and thus the rank of this Jacobian is still k . Since the first k columns are linearly independent, this means that the remaining $n - k$ must be 0. That is, each A^i is independent

of y , since its partial derivatives with respect to each y^j is zero (in other words, $A(x, \bullet)$ is constant). So $A(x, y) = A(x, 0)$ and thus

$$F \circ \varphi^{-1}(x, y) = (x, A(x, y)) = (x, A(x, 0)) = F \circ \varphi^{-1}(x, 0),$$

so φ^{-1} defines the desired chart.

- (2) Notice that for a projection P , its rank is just its trace. This is because if it projects onto V parallel to W , taking bases of V and W , P 's representation relative to the union of this base is $\text{diag}(1, \dots, 1, 0, \dots, 0)$, whose trace is its rank. Trace is a continuous function $\text{Mat}_n(\mathbb{R}) \rightarrow \mathbb{R}$, and thus the rank of $A(t)$ depends continuously on t . The set of ranks of $A(t)$ is a set of integers, which is disconnected when not a singleton. The set of ranks is the continuous image of $T \rightarrow A(t) \rightarrow \text{tr}A(t)$ and since T is connected, so must the set of traces be, thus it is a singleton, as desired.
- (3) For $q \in \text{im}r$, we have that since $r \circ r = r$,

$$dr_q = d(r \circ r)_q = dr_{rq} \circ dr_q = dr_q \circ dr_q.$$

Thus dr_q is a projection, and the map $q \rightarrow dr_q$ is continuous so that by above dr_q 's rank is constant (since $\text{im}r$ is the continuous image of a connected space). Let this constant rank be k , then note it is also r 's maximum rank: for all $q \in M$ we have that $dr_q = dr_{rq} \circ dr_q$, and since composition does not increase rank, dr_q 's rank is bound by $dr_{rq} = k$.

Let Q be the set of all points $q \in M$ where dr_q 's rank is equal to k . This contains $\text{im}r$ as already shown, but it is also open. Indeed, for $q \in Q$, dr_q has rank k and in a suitable neighborhood of q the rank of r remains k .

This is because we can look at dr 's matrix representation relative to open charts around q and rq (i.e. the Jacobian of r 's coordinate representation). Since dr_q has rank k , the matrix has an invertible $k \times k$ submatrix. Consider then the function which maps a point p (which lies in the chart around q) to the determinant of this $k \times k$ submatrix of dr_p . If a point p is mapped to a nonzero value, then this $k \times k$ submatrix of dr_p is invertible thus dr has rank $\geq k$ at p ; since k is the maximal rank, dr_p 's rank is equal to k . So the preimage of the open set $\mathbb{R} - 0$ is a neighborhood of p contained in Q , so Q is open.

- (4) Our first point only required that F have constant rank in a neighborhood, which is true here (the neighborhood is Q). Thus there is $(\mathbf{x}, U)$ with the desired property, but not necessarily $r(\mathbf{x}(U)) \subseteq \mathbf{x}(U)$.

We want $r(\mathbf{x}(u)) \in \mathbf{x}(U)$ for all $u \in U$, i.e. that $u \in (r \circ \mathbf{x})^{-1}(\mathbf{x}(U))$, which is open. Thus intersecting $(r \circ \mathbf{x})^{-1}(\mathbf{x}(U))$ with U and restricting $\mathbf{x}$ gives us the property. Indeed,

$$r(\mathbf{x}((r \circ \mathbf{x})^{-1}(\mathbf{x}(U)) \cap U)) \subseteq r(\mathbf{x}(U)) \cap \mathbf{x}(U) = \mathbf{x}((r \circ \mathbf{x})^{-1}(\mathbf{x}(U)) \cap U).$$

- (5) Clearly we have that $r(M) \cap \mathbf{x}(U) \subseteq r(\mathbf{x}(U)) \cap \mathbf{x}(U)$. Conversely if $r(p) \in \mathbf{x}(U)$ then $r(p) = r(r(p)) \in r(\mathbf{x}(U))$, as desired. Since $r(\mathbf{x}(U)) \subseteq \mathbf{x}(U)$, we have that $r(M) \cap \mathbf{x}(U) = r(\mathbf{x}(U))$ even.
- (6) Consider a chart $(U, \mathbf{x})$ as above, then we can make $r(M) \cap \mathbf{x}(U) = r(\mathbf{x}(U))$ an embedded k -manifold. Indeed, consider the projection $\pi: \mathbb{R}^n \rightarrow \mathbb{R}^k$ and inclusion by concatenating zeroes, $\iota: \mathbb{R}^k \rightarrow \mathbb{R}^n$. So we have $r \circ \mathbf{x} = r \circ \mathbf{x} \circ \iota \circ \pi$ by assumption. Define the chart $(U', \mathbf{y})$ with $U' = \pi(U) \subseteq \mathbb{R}^k$ by $\mathbf{y} = r \circ \mathbf{x} \circ \iota$, i.e. $\mathbf{y}(x) = r \circ \mathbf{x}(x, 0)$. This is surjective by construction. Now, we know that from our first point, $r \circ \mathbf{x}(x, y) = (x, R(x))$ for some function $R: \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$. Thus $\mathbf{y}$ has the form $x \mapsto (x, R(x))$ and is thus clearly injective. Its inverse is $(x, y) \mapsto x$, which is clearly smooth thus $\mathbf{y}$ is a diffeomorphism.

It remains to show that $\mathbf{y}(U')$ is open in $r(M)$. Notice that since $r \circ \mathbf{x}(x, 0) = r \circ \mathbf{x}(x, y)$,

$$\mathbf{y}(U') = \{r \circ \mathbf{x}(x, 0) \mid x \in U'\} = \{r \circ \mathbf{x}(x, y) \mid (x, y) \in U\} = r(\mathbf{x}(U)) = r(M) \cap \mathbf{x}(U),$$

which is open in the subspace topology. And for $\mathbf{y}_1$ and $\mathbf{y}_2$ corresponding to $\mathbf{x}_1, \mathbf{x}_2$, we note that $\mathbf{y}_1^{-1} \circ \mathbf{y}_2$ is just the identity. As we said, $\mathbf{y}$ has the form $x \mapsto (x, R(x))$ with inverse $(x, y) \mapsto x$, so $\mathbf{y}_1^{-1} \circ \mathbf{y}_2$ maps x to x , so this does define a smooth atlas on $r(M)$.

To show that $r(M) \subseteq M$ is a submanifold, we need to show that the inclusion is an immersion. Let $\iota: r(M) \hookrightarrow M$ be the inclusion, then relative to $(U, \mathbf{x})$ and $(U', \mathbf{y})$ (since $\mathbf{y}(U') \subseteq \mathbf{x}(U)$) it is equal to

$$\mathbf{x}^{-1} \circ \iota \circ \mathbf{y}(x) = \mathbf{x}^{-1} \circ r \circ \mathbf{x}(x, 0),$$

which is smooth and has constant rank k since r is and does, and U' 's dimension is k so this is indeed a smooth immersion. Since $r(M)$ has the subspace topology, the inclusion is a smooth embedding, as desired.

3.4 Problem

Consider the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$. Let $A: S^n \rightarrow S^n$ be the antipodal map $p \mapsto -p$. Using any oriented atlas of S^n , show that A is orientation preserving iff n is odd.

Consider the stereographic projections at e_1 and $-e_1$, denoted by φ_+ and φ_- , $S^n - \{\pm e_1\} \rightarrow \mathbb{R}^n$. These map

$$\varphi_+(x_0, \dots, x_n) = \frac{1}{1 - x_0}(x_1, \dots, x_n), \quad \varphi_-(x_0, \dots, x_n) = \frac{1}{1 + x_0}(x_1, \dots, x_n).$$

These are restrictions of the obviously smooth maps $\{x \in \mathbb{R}^{n+1} \mid x_0 \neq \pm 1\} \rightarrow \mathbb{R}^n$, and since S^n is a submanifold, the restrictions are smooth as well. For $(X_1, \dots, X_n) \in \mathbb{R}^n$, let $N^2 = |\bar{X}|^2 = \sum_i X_i^2$, then their inverses are given by

$$\varphi_+^{-1}(X_1, \dots, X_n) = \frac{1}{N^2 + 1}(N^2 - 1, 2X_1, \dots, 2X_n), \quad \varphi_-^{-1}(X_1, \dots, X_n) = \frac{1}{N^2 + 1}(1 - N^2, 2X_1, \dots, 2X_n).$$

These again are smooth, as the restrictions of the obviously smooth maps $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$, and again we can restrict the codomain to S^n since it is a submanifold.

The transition map is

$$\varphi_- \circ \varphi_+^{-1}(X) = \frac{X}{|X|^2},$$

and its Jacobian can be computed:

$$J_{ij} = \begin{cases} -\frac{2X_i X_j}{|X|^4} & i \neq j \\ \frac{|X|^2 - 2X_i^2}{|X|^4} & i = j \end{cases}.$$

This has negative determinant: for $X = (1, 0, \dots, 0)$ it is $\text{diag}(-1, 1, \dots, 1)$.

In lecture we showed that this means we can replace φ_- with $r \circ \varphi_-$ where r is the map which flips the sign of the first coordinate, and obtain an oriented atlas. For the antipodal map A , notice that $\varphi_- = A \circ \varphi_+ \circ A$, and thus A 's coordinate representation is

$$r \circ \varphi_- \circ A \circ \varphi_+^{-1} = r \circ A \circ \varphi_+ \circ \varphi_+^{-1} = r \circ A,$$

since $A^2 = \text{id}$. This is the map $X \mapsto (X_1, -X_2, \dots, -X_n)$, whose Jacobian is $\text{diag}(1, -1, \dots, -1)$ which has determinant $(-1)^{n-1}$. This is positive only when n is odd, as desired. For $\varphi_+ \circ A \circ \varphi_-^{-1}$ we get the same result by symmetry.

3.5 Problem

Let S^{2n+1} be the unit sphere in $\mathbb{C}^{n+1}$. Define an action of S^1 on S^{2n+1} by

$$z(w^1, \dots, w^{n+1}) = (zw^1, \dots, zw^{n+1}).$$

(1) Show that this action is smooth.

(2) Show that this action is free and proper, and that the orbit space S^{2n+1}/S^1 is diffeomorphic to $\mathbb{CP}^n$.

- (1) This action is just the restriction to $S^1 \times S^{2n+1} \rightarrow S^{n+1}$ of the map $\mathbb{C} \times \mathbb{C}^{n+1} \rightarrow \mathbb{C}^{n+1}$ given by scalar multiplication:

$$z(w^1, \dots, w^{n+1}) = (zw^1, \dots, zw^{n+1}).$$

This is a smooth map, and $S^1 \subseteq \mathbb{C}$ and $S^{2n+1} \subseteq \mathbb{C}^{n+1}$ are embedded submanifolds, thus the restriction of scalar multiplication to $S^1 \times S^{2n+1} \rightarrow S^{2n+1}$ remains smooth, by our above proof.

- (2) The action is free since if $z(w^1, \dots, w^{n+1}) = (w^1, \dots, w^{n+1})$ then since w is a point in S^{2n+1} it must have some nonzero coordinate, say w^i . Then $zw^i = w^i$ implies that $z = 1$.

The action is proper since S^1 is compact. Since if (z_i) is any sequence in S^1 , it has a convergent subsequence by virtue of S^1 being compact (without even considering a sequence (w_i) of S^{2n+1}).

Consider the quotient map $\pi: \mathbb{C}^{n+1} - 0 \rightarrow \mathbb{CP}^n$ mapping w to the linear subspace spanned by it, $\pi(w) = [w]$. Since $S^{2n+1} \subseteq \mathbb{C}^{n+1} - 0$ is an embedded submanifold, we can restrict this to $\pi: S^{2n+1} \rightarrow \mathbb{CP}^n$. Define S^1 to act on $\mathbb{CP}^n$ trivially: $z[v] = [v]$ for all $[v] \in \mathbb{CP}^n$. Then π is S^1 -equivariant, since $\pi(zw) = [zw] = [w] = z[w]$, and thus has constant rank. Furthermore, π is surjective (any vector can be normalized), and thus it must be a smooth submersion. This is because locally π has the coordinate representation $(x_1, \dots, x_n) \rightarrow (x_1, \dots, x_k, 0, \dots, 0)$ which is a surjection only when the rank k is full.

And also because π is S^1 -equivariant, it quotients to $\pi/S^1: S^{2n+1}/S^1 \rightarrow \mathbb{CP}^n$ given by $S^1 \cdot w \mapsto \pi(w)$. Notice that the following diagram commutes:

$$\begin{array}{ccc} & S^{2n+1}/S^1 & \\ \pi_1 \nearrow & & \searrow \pi/S^1 \\ S^{2n+1} & \xrightarrow{\pi} & \mathbb{CP}^n \end{array}$$

where π_1 is the covering map of the quotient manifold. Since $\pi = (\pi/S^1) \circ \pi_1$ is a submersion, we have that π/S^1 must be a submersion as well: for any $w \in S^{2n+1}$ we have that

$$d\pi_w = d(\pi/S^1)_{\pi_1(w)} \circ d(\pi_1)_w$$

is surjective, and thus so too must $d(\pi/S^1)_{\pi_1(w)}$. Since π_1 is surjective, this means π/S^1 is a submersion.

But π/S^1 is also bijective: as the quotient of the surjective π , it remains surjective. And if $\pi(w) = \pi(u)$ then $w = zu$ for some $z \in \mathbb{C}$, and since both w, u have norm 1 this means $z \in S^1$. Thus $S^1 \cdot w = S^1 \cdot u$ so that π/S^1 is injective. A bijective submersion is a diffeomorphism (it has the local representation $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_k)$ and since it is bijective we must have that $n = k$, so that it is a local diffeomorphism, and bijective local diffeomorphisms are global diffeomorphisms).

So we have shown that $\pi/S^1: S^{2n+1}/S^1 \rightarrow \mathbb{CP}^n$ is a diffeomorphism, as desired.